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Untargeted LC-MS/MS metabolomics reveals farming practice- and cultivar-driven modulations of pea (*Pisum sativum* L.) seed metabolome across multiple biosynthetic pathways

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ABSTRACT

This work presents a comprehensive biochemical landscape of pea seeds captured by untargeted LC-MS-based metabolomics of ten pea cultivars grown at three Danish field sites following different agricultural practices. More than 1200 metabolite features were detected in methanolic extracts of pea seed flours. Of these, nearly 300 features were identified using mass spectral libraries and advanced computational tools. Approximately 40 metabolites were found to be associated with location effect, independent of cultivar type. Organically grown pea samples showed lower levels of the main pea triterpene glycoside soyasaponin I and higher levels of nitrogen-abundant amino acids, indicating increased nitrogen availability in soil. More than 100 metabolites were associated with the location-independent cultivar effect. Akooma and Greenway cultivars showed the most distinct metabolome with greater levels of polyunsaturated fatty acids and lipid oxidation products known to give 'beany' off-flavors. The commonly cultivated pea variety, Ingrid, was devoid of compounds derived from the phenylpropanoid pathway including hydroxycinnamic acid amides such as caffeoyl, feruloyl, and coumaroyl aspartates that were present in all other cultivars. Three chloroauxin metabolites, reported here for the first time, were identified through molecular networking within GNPS platform and propagation of annotation from a computationally predicted indole-3-acetic acid catabolite. Overall, the results indicate biochemical adaptation of pea plants to location or agricultural practices as reflected in their seed metabolome.

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1. Introduction

The awareness of sustainable food production is increasing due to the growing global population and climate change, along with a rising recognition of the importance of a balanced diet for human health [1]. Considering these factors, more attention is being directed toward expanding the research on various crops to diversify dietary options while promoting both health and environmental resilience [2]. Besides the major staple crops such as wheat, rice, maize, and soybean, the pea seeds (*Pisum sativum* L.) provide notable agricultural and nutritional benefits. Although the starch content of dried pea seeds ranges between 40% and 60%, their glycemic index is relatively lower than that of most carbohydrate-rich crops, making them a calorific yet nutritionally healthier option [3]. As members of the nitrogen-fixing *Fabaceae* family, pea seeds are rich in protein and comprise an overall average of 23.4% of their dry mass [3]. The lipid content of peas ranges from 0.9% to 5.0% and the amount of polyunsaturated fatty acids (PUFA) is higher than that of saturated fatty acids [4].

As sessile organisms, plants have developed a remarkable genetic capacity to produce a diverse array of specialized metabolites that enable them to adapt to environmental challenges [5]. While these small molecules serve critical roles in plant defense and adaptation, they are also of significant interest in the context of plant-based foods due to their effects on nutritional value, sensory properties, and potential health benefits. Several glycosides of quercetin, apigenin, luteolin, kaempferol, myricetin, and isorhamnetin have been identified in pea seeds and are known to contribute to their antioxidant and anti-inflammatory properties [6,7]. Like many legumes, pea seeds contain various triterpenoid saponins, such as group B saponins (soyasaponin Ba, Bb, Bx, Bc) and DDMP (2,3-dihydro-2,5-dihydroxy-6-methyl-4H-pyran-4-one) saponins (soyasaponin αg, βg, γg, αa, βa), which are considered

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to account for their bitterness and astringency [8]. Despite the concentrations of these secondary metabolites are relatively small in pea seeds, they play a crucial role in the digestibility of the dominating macromolecules like pea protein and fiber and contribute to the overall nutritional value of pea [9]. It is therefore important to map not only macromolecules but also small molecular metabolites in pea, shaping its chemical space to better understand its nutritional value, sensory characteristics, and potential impact on human health. Moreover, the variation in cultivar type and growing conditions, including geographical location and agricultural practices, influences biosynthetic pathways in pea, resulting in differences in their metabolome. Studies have shown variation in the profiles of amino acids, lipid-like molecules, polyketides, and phenylpropanoids in eight pea cultivars grown in different regions of Spain using the combination of targeted and untargeted LC-MS metabolomics [10]. In a study of two pea cultivars grown under normal conditions and biotic stress represented as contamination with soil-borne pathogens, cultivar effect was shown to influence volatile compounds and lipoxygenase activity [11]. This enzyme leads to lipid peroxidation of PUFAs as a response to pathogen attack, which eventually contributes to the accumulation of off-flavor substances in pea seeds.

Several analytical methods have been used to examine the chemical profile of crops, both qualitatively and quantitatively [12–15]. In this regard, untargeted metabolomics facilitates an unbiased analysis of all detectable metabolites with molecular masses ≤ 1500 Da in biological systems. Combined with chemometrics, it uncovers distinct molecular patterns associated with one or more factors included in a study design [14]. Ultra-high performance liquid chromatography (UPLC) coupled with high-resolution mass spectrometry (HRMS) has become the method of choice in metabolomics due to its broad metabolite coverage, high sensitivity, and selectivity. When combined with emerging computational mass spectrometry approaches [16,17] liquid chromatography–tandem mass spectrometry (LC-MS/MS) enables comprehensive profiling of small molecules in biological systems, making it a powerful tool in metabolomics and foodomics research [18,19]. Moreover, specific methods such as widely targeted metabolite modificomics have been developed that utilizes modification-specific neutral loss scanning to enable large-scale, high-confidence detection of modified metabolites, expanding the chemical coverage beyond what untargeted metabolomics can achieve [20]. Other untargeted metabolomics data mining tools such as MassQL [21], which enables the querying local and repository-scale intrinsic mass spectrometry data patterns, and MS2LDA [22], which identifies recurrent fragment and neutral-loss motifs corresponding to molecular substructures, further facilitate the detection of diverse metabolic signatures.

In this study, untargeted LC-MS/MS-based metabolomics was employed to investigate the metabolite variation in pea seed flours from ten cultivars grown across three geographic locations in Denmark, including two farming sites managed conventionally and one site managed under organic agricultural practices (Fig. 1). Mass spectral libraries along with mass spectral computational tools for molecular networking in Global Natural Products Social Molecular Networking (GNPS) platform paired with molecular structure identification in SIRIUS were employed to annotate metabolites in LC-MS/MS data. Using the propagation of annotation in molecular networking within the GNPS platform, the structure of three chloroauxins is putatively annotated for the first time. Partitioning of the metabolomics data by ANOVA-Simultaneous Component Analysis (ASCA) revealed the distinct effect of growth location and pea cultivar, and their interaction on pea seed metabolome.

2. Materials and methods

2.1. Field trials and samples

Ten cultivars of field peas (*Pisum sativum* L.), Akooma, Astronaute, Bagoo, Greenway, Ingrid, Kaplan, Karacter, Manager, Orchestra, and Skol, were grown at three different geographic locations in 2022 (10 plots per location, i.e., 30 samples in total). At two locations (Sejet and Gamborg), peas were grown according to conventional principles. Sejet is a clay loam/heavy clay soil, while Gamborg is a silty loam/clay loam soil. Within each location, the trial was physically placed to reduce impacts of soil heterogeneity. Peas were sown on April 1st or March 31st at the Sejet and Gamborg locations, respectively. Peas at the Gamborg location received 400 kg 0-4-21 synthetic NPK fertilizer per hectare at sowing. At the Sejet and Gamborg locations, pesticides were applied to control weeds, fungi, and pests. At the third location (Hammel, a coarse sandy clay soil), peas were grown according to the principles of organic farming (EU Regulation 2018/848). Sowing was conducted on the 12th of May and no fertilizers or pesticides were applied. At maturity, ultimo August/primo September, peas were harvested and stored in aerated bags.

2.2. Sample processing

All samples were extracted in 80% methanol (HPLC-grade, Sigma-Aldrich, Germany) in randomized order in 2 mL Eppendorf tubes. The extraction solvent contained 4 μmol L^{-1} sulfadimethoxine (VETRANAL, analytical standard, Sigma-Aldrich, Germany) as an internal standard to monitor LC-MS/MS system performance. Samples were vortexed (Vortex-Genie 2, Scientific Industries, Bohemia, NY, USA) for 10 s, followed by ultrasonication (Sonorex Digitec DT 100H, Germany) at 35–40 °C for 30 min. Samples were vortexed again for 10 s and centrifuged (ScanSpeed 1580R, LaboGene, Denmark) at 21,124×*g* for 20 min at 4 °C and the supernatant was analyzed.

2.3. LC-MS/MS data acquisition

Extracted samples were randomized and analyzed using an Elute UPLC system coupled with an impact II QTOF-MS (Bruker Daltonik, Germany). Pooled quality control and blank samples were analyzed every eight biological samples. Chromatographic separation was achieved using a reverse-phase Intensity Solo 2 C18 column (100 Å, 2 μm, 100 mm × 2.1 mm, Bruker Daltonik) preceded by an ACQUITY UPLC BEH C18 VanGuard pre-column (130 Å, 1.7 μm, 2.1 mm × 5 mm, Waters Corporation, MA, USA), maintained at 40 °C. The mobile phase consisted of solvent A (Milli-Q water + 0.1% formic acid, Merck, Germany) and solvent B (acetonitrile + 0.1% formic acid, hypergrade LiChrosolv, Merck, Germany) with a constant flow rate of 0.3 mL min^{-1}. The gradient elution was as follows: 0–2 min (100% A), 2–8 min (linear gradient to 100% B), 8–11 min (100% B), 11–11.01 min (return to 0% B), and 11.01–14 min (re-equilibration at 0% B).

Mass spectra were recorded from *m*/*z* 20 to 1300 at a spectral rate of 12 Hz using electrospray ionization (ESI) in positive-ion mode. The instrument was externally calibrated using a 10 mmol L^{-1} sodium formate solution in 50% isopropanol before sample analysis and tuned with ESI-L Low Concentration Tuning Mix (Agilent Technologies, CA, USA). ESI conditions were: end plate offset - 500 V, capillary voltage - 4500 V, nebulizer pressure - 2.2 bar, dry gas flow - 10 L min^{-1}, and drying temperature - 220 °C. MS conditions were: funnel 1 and 2 RF - 200 Vpp, hexapole - 50 Vpp, ion energy - 5 eV, collision energy - 5 eV, pre-pulse storage - 5 μs. Tandem mass spectra were acquired in data-dependent acquisition

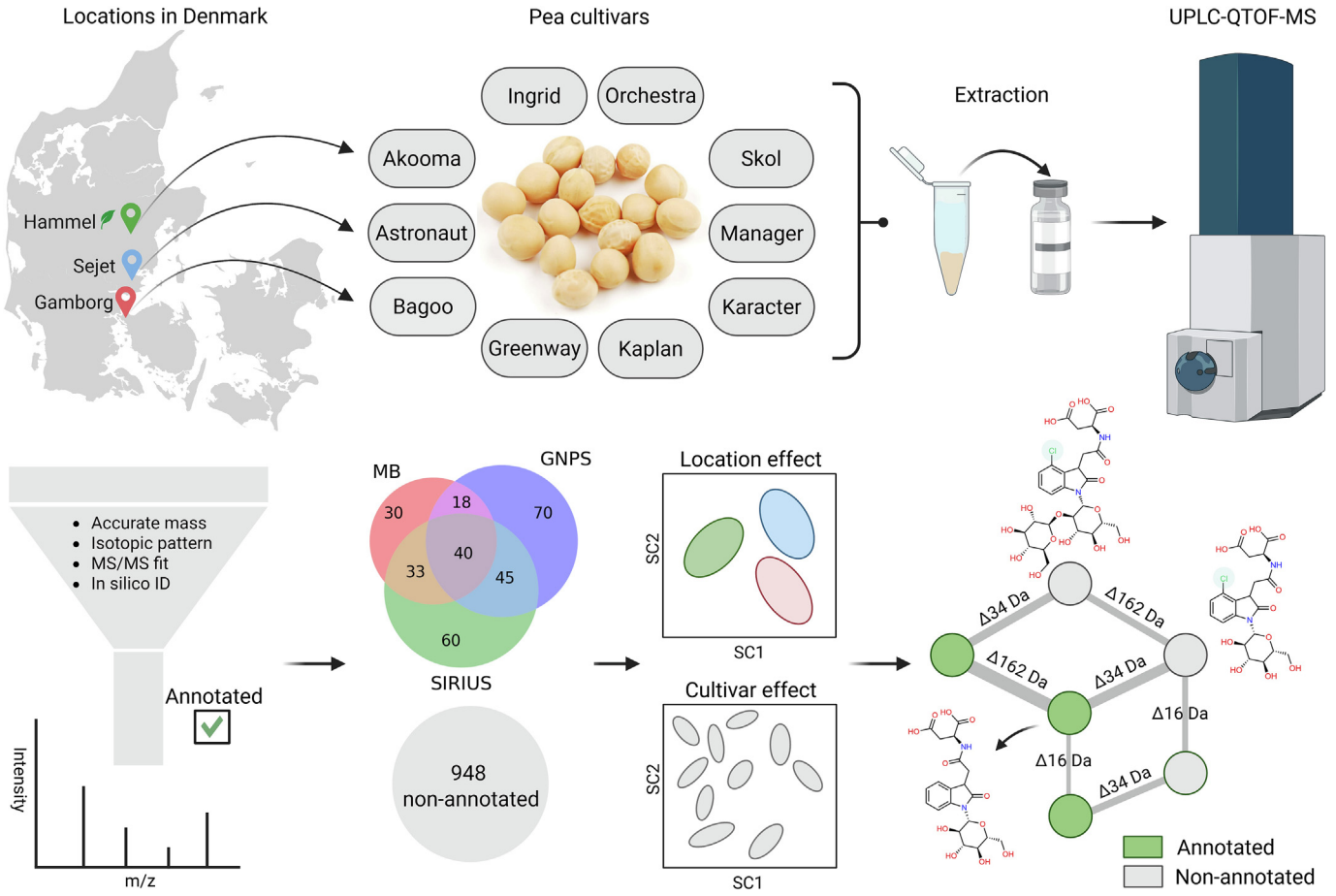


Fig. 1. UPLC-QTOF-MS-based untargeted metabolomics study of ten pea cultivars grown in three different locations in Denmark with organic agricultural practices used in Hammel. In total, 90 samples (10 cultivars × 3 locations × 3 replicates) along with pooled quality control samples were extracted in 80% methanol with addition of internal standard to acquire untargeted MS data. Raw data were processed and annotated using spectral libraries and computational approaches with pre-defined parameters, which was then analyzed using ANOVA-simultaneous component analysis (ASCA) to reveal the effect of location, cultivar, and their interaction on pea metabolome. The putative annotation of significant non-annotated features was additionally revealed using computational approaches GNPS and SIRIUS. Created in BioRender.com.

mode, selecting three precursor ions per scan cycle, with active exclusion after three spectra and a release period of 0.20 min.

2.4. Data analysis

Feature extraction was performed with MetaboScape 2021b (Bruker Daltonik): minimum features in ≥ 2 analyses, minimum presence in ≥ 3 analyses, intensity threshold of 5000 counts, and a minimum peak length of 6 spectra (4 spectra for recursive extraction). The total ion chromatograms (TICs) of QC samples were inspected and showed stable signal intensity throughout the data acquisition, indicating negligible signal drift. Normalization against the internal standard added to the extraction solvent did not result in notable data quality improvement reflected in RSD values of features across QC samples. Spectral annotation employed hierarchical search mode with precursor *m/z* tolerances of 2–5 ppm, mSigma tolerances of 20–50, and MS/MS scores ranging from 900 to 600. Feature-based molecular networking (v2024.10.09) was conducted within the GNPS2 ecosystem applying a precursor ion tolerance of 0.05 Da, fragment ion tolerance of 0.05 Da, cosine score, and minimum matched peaks set at 0.65 and 6, respectively. Additional annotation of features was performed in SIRIUS platform (v.6.3) and putative candidates with COSMIC confidence score ≥ 0.4 were filtered for statistical analysis. Conflicting or overlapping annotations across platforms were resolved through a hierarchical approach, prioritizing (1) spectral library-based matches in MetaboScape and GNPS, (2) SIRIUS predictions confirmed by manual spectral interpretation supported by literature evidence. The complete feature table with annotations from all annotation platforms used in this study is given in Table S1. ANOVA-simultaneous component analysis (ASCA) [23] was performed in MATLAB ver. R2023a (9.14.0.2337262) on autoscaled data using MATLAB scripts written by the authors. A one-way ANOVA was performed with p-values adjusted for multiple testing using the Benjamini-Hochberg procedure (FDR = 5%) in python (v.3.6) using scipy and statsmodels libraries. Hierarchical clustering heatmaps were generated using Ward's method and Euclidean distance on row-wise max-normalized data using scipy library. The matplotlib and seaborn libraries were used for visualization of the plots.

3. Results

3.1. Pea metabolome

The final feature table included 1230 variables after Metaboscape-based processing of raw LC-MS/MS data files obtained on 90 pea samples (3 locations × 10 cultivars × 3 replicates). Using mass spectral libraries implemented in Metaboscape, 121 features were identified at Level 2 in accordance with the Metabolomics Standards Initiative [24]. Similarly, feature-based molecular networking (FBMN) using publicly available spectral

libraries within the GNPS platform allowed annotation of 173 features [25]. 118 GNPS annotations were derived from the nearest-neighbor suspect spectral library, which was generated through the propagation of annotations of reference compounds in repository-scale molecular networking [26]. Additionally, 178 metabolites with COSMIC confidence score ≥ 0.4 were annotated using molecular formula and structure identification tool CSI (Compound Structure Identification): FingerID [27,28] in SIRIUS platform. The Venn diagram in Fig. 1 shows the distribution of metabolite annotations across all three sources. Further, compound class prediction using CANOPUS in SIRIUS platform was performed to assign molecular class to 424 metabolites representing 42 molecular classes [29]. Major molecular classes included 126 fatty acid derivatives, 74 amino acids, small peptides, and oligopeptides, 71 glycero(phospho) lipids, 44 alkaloids, 23 terpenoids, 17 nucleosides, 11 phenylpropanoids (C6–C3), 9 flavonoids and phenolic acids, and 9 saccharides. The downstream data analysis was performed separately (1) for 121 metabolites annotated in Level 2 using Metaboscape, (2) for 178 metabolites annotated in Level 3 using SIRIUS, and (3) for all 1230 variables, including 948 unknown features. Table 1 summarizes the selected features influenced by location and cultivar effect by providing their *m*/*z* and retention time values, annotation confidence level, effect size, and *P*-value.

3.2. ASCA-based partitioning of the variation in data according to the study design

The full-factorial balanced design of the study, including two factors: 1) location with three levels and 2) cultivar with ten levels, allowed the use of ASCA with permutation testing (n = 1000) to investigate the variation present in the data that derive from the two factors, and their interaction terms on pea metabolome (Fig. 2). Both factors, location and cultivar, and their interaction effect were shown to be statistically significant when considering the 121 metabolites identified on the Level 2 annotation confidence, with the highest variation attributed to the cultivar effect (28.6%, P = 0.001), followed by the two-factor interaction effect explaining 17.2% of variation (P = 0.02). The location effect explained the lowest variation of 7.8% (P = 0.001) in the Level 2 metabolome (121 metabolites).

3.3. Effect of location

As illustrated in the ASCA scores plot (Fig. 2A), the main variation attributed to the location effect was driven by metabolome differences between the samples from Hammel and those from the other two sites, Gamborg and Sejet. Simultaneous component 1 (SC1) primarilly differentiates the Hammel samples from those grown in Gamborg and Sejet, while SC2 differentiates the Gamborg and Sejet samples. Given that peas in Hammel were cultivated organically, whereas those in Gamborg and Sejet were conventionally grown, the clear separation along SC1 could be associated with metabolome differences arising from agricultural practices. In this regard, ASCA loadings plot demonstrated the distinctive patterns of metabolites distinguishing the samples from the three locations (Fig. 2B). Specifically, the metabolite profile distinguishing Hammel samples included increased levels of several amino acids—pipecolic acid, histidine, asparagine, homoarginine, and homocitrulline — as well as lipid-like compounds such as 13-keto-9, 11-octadecadienoic acid, 1-pentadecanoyl-*sn*-glycero-3-phosphocholine, and pantothenic acid. In contrast, Gamborg samples were characterized by higher levels of adenosine monophosphate (AMP), 2-deoxyadenosine, and glycerophosphocholine, while Sejet samples were richer in N-phenylacetylaspartic acid, choline, and LysoPC (14:0/0:0).

The ASCA analysis performed on the Level 3 identified metabolites, including 178 metabolites annotated via SIRIUS (Fig. S1A), revealed a similar trend in the location effect (7.3%, P = 0.001). Among the computationally annotated metabolites differentiating the Hammel samples from those of the other two locations were pisatin, a common pea phytoalexin, argininosuccinic acid, and several fatty acids. Sejet-grown peas exhibited relatively higher levels of arginine-C5:0 (Arg-C5:0), N-salicyloylaspartic acid, and dimethylarginine, while the Gamborg samples were characterized by elevated levels of triterpenoid saponins, including dehydrosoyasaponin I and soyasaponin I (Fig. S1B). The ASCA scores plot of all 1230 features depicted the partial separation of Hammel samples from those grown in Sejet and Gamborg (6.4%, P = 0.001) (Fig. S2A). Notably, samples from all three locations were split into two subpopulations, differentiating along SC1 (73%) and SC2 (27%). Metabolites such as AMP, pipecolic acid, Arg-C5:0, and N-phenylacetylaspartic acid contributed notably to the differentiation of location types (Fig. S2B).

To assess how individual metabolites vary across the three locations, independent of cultivar types, a one-way ANOVA was performed with Benjamini-Hochberg correction for multiple testing (FDR = 5%). Of the 1230 features analyzed, 205 showed significant differences in levels across locations. Among these, 33 out of 121 Level-2 identified metabolites differed significantly among pea seeds from the three locations. Pipecolic acid exhibited the largest effect size (η^2) of 47.8% (P = 3.89e-11), being more abundant in Hammel samples compared to the other two conventional farming sites, Gamborg and Sejet (Fig. 3B). Homoarginine (η^2 = 36.6%, P = 1.28e-07), homocitrulline (η^2 = 32.4%, P = 1.28e-07), and spermidine (η^2 = 31.1%, P = 2.02e-06) showed a similar trend, being most abundant in Hammel, followed by Sejet and Gamborg samples. Conversely, N-phenylacetylaspartic acid (η^2 = 35%, P = 2.71e-07) was most abundant in Sejet samples, followed by Hammel and Gamborg, while adenosine 5-monophosphate (η^2 = 32%, P = 1.31e-06) demonstrated the reverse trend. The remaining significant metabolites showed effect sizes ranging from 10 to 19%. The 33 metabolites that significantly differ between the three locations were visualized using a hierarchically clustered heatmap (Fig. 3A). Hierarchical clustering revealed two main clusters: one comprising metabolites predominantly abundant in Hammel samples and another showing metabolites abundant in Gamborg and Sejet samples. Notably, amino acids with elevated abundance in Hammel such as homocitrulline, asparagine, homoarginine, and polyamine spermidine exhibited similar intensity patterns and were combined into one group.

Among the 178 metabolites annotated using SIRIUS, a one-way ANOVA identified 41 metabolites that are significantly influenced by location (FDR-adjusted P-values < 0.05) (Fig. S3B). Of these, argininosuccinic acid, a precursor to arginine in plants, exhibited the largest effect size of 52.1% (P = 2.31e-12), with the highest abundance in Hammel samples, followed by Sejet and Gamborg. Pisatin (η^2 = 41.5%, P = 2.62e-09) showed a similar pattern. Additionally, the feature annotated as C5:0-Arg had an effect size of 42.8% (P = 1.22e-09) and was slightly more abundant in Sejet samples compared to Hammel, with Gamborg samples showing the lowest levels. Pisumsaponin I (η^2 = 10.7%, P = 3.29e-02) showed marginally higher levels in Gamborg and Hammel samples. Interestingly, soyasaponin I (η^2 = 11.7%, P = 2.36e-02) showed its lowest abundance in Hammel samples, contrary to other secondary metabolites known for plant defense mechanisms, such as pisatin and pisumsaponin I.

When visualizing all location-influenced SIRIUS annotated metabolites in a heatmap, two distinct clusters emerged. The first cluster included 21 metabolites predominantly abundant in Gamborg and Sejet samples compared to samples from the organic farming site at Hammel. The second cluster included 17 metabo-

Table 1
Metabolites showing significant difference between pea cultivars and/or between pea plants grown in three different locations.

Metabolite name	Rt (min)	*m/z*	Adduct	Annotation level	Δ*m/z* (mDa)	mSigma	MSMS score	Cosine score	Confidence score	Location/ Cultivar	*P*-value	Effect size (%)
Location effect												
Pipecolic acid	1.02	130.0863	$[M + H]^+$	2	0.007	2.9	813.6	–	–	Hammel↑	3.07e-11	47.8
Homocitrulline	0.99	190.1184	$[M + H]^+$	2, 3	0.239	6.5	749.1	–	0.943	Hammel↑	9.65e-07	32.4
Asparagine	0.90	133.0605	$[M + H]^+$	2, 3	0.239	5.4	978.9	0.896	0.993	Hammel↑	1.32e-03	18.9
Homoarginine	0.93	189.1342	$[M + H]^+$	2, 3	0.435	8.5	984.4	0.939	0.921	Hammel↑	1.01e-07	36.6
Spermidine	0.71	146.165	$[M + H]^+$	2, 3	0.219	6.9	711.9	–	0.716	Hammel↑	1.59e-06	31.1
Adenosine monophosphate	1.00	348.07	$[M + H]^+$	2, 3	0.374	10.3	994.7	–	0.936	Gamborg↑	1.03e-06	32.0
Deoxyadenosine	3.93	268.1036	$[M + H]^+$	2, 3	0.453	11.0	991.2	0.963	0.949	Gamborg↑	8.23e-03	14.3
Uridine	1.05	245.0764	$[M + H]^+$	2, 3	0.448	9.8	889.4	–	0.927	Hammel↓	3.11e-06	29.8
Homopipecolic acid	1.00	144.1017	$[M + H]^+$	3	–	–	–	–	0.444	Hammel↑	5.08e-10	44.4
Argininosuccinic acid	0.99	291.1295	$[M + H]^+$	3	–	–	–	–	0.543	Hammel↑	2.31e-12	52.1
Arg-C5:0	4.30	259.1761	$[M + H]^+$	2,3	–	–	–	0.84	0.463	Sejet↑	1.22e-09	42.8
Niacin	1.18	124.0391	$[M + H]^+$	3	–	–	–	–	0.845	Sejet↑	6.64e-06	28.6
Pisatin	6.61	297.0754	$[M+H–H_2O]^+$	2,3	–	–	–	0.97	0.546	Hammel↑	2.62e-09	41.5
N-salicyloyl aspartic acid	4.31	254.0645	$[M + H]^+$	3	–	–	–	–	0.767	Sejet↑	1.06e-03	19.3
N-phenylacetyl aspartic acid	4.89	252.0862	$[M + H]^+$	2, 3	0.402	4.2	795.7	–	0.962	Sejet↑	2.14e-07	35.0
Cultivar effect												
Chromone	4.58	147.0437	$[M + H]^+$	2	0.326	10.0	766.7	–	–	Ingrid↓	3.87e-39	92.5
Protocatehuic acid	4.01	155.0335	$[M + H]^+$	2, 3	0.366	7.5	752.2	–	0.763	Ingrid↓	7.74e-23	80.1
N-caffeoylaspartic acid	4.37	296.0762	$[M + H]^+$	3	–	–	–	0.917	0.538	Ingrid↓	7.61e-51	96.2
N-coumaroylaspartic acid	4.59	280.0812	$[M + H]^+$	3	–	–	–	0.959	0.478	Ingrid↓	2.54e-42	93.7
N-feruloylaspartic acid	4.70	310.0918	$[M + H]^+$	3	–	–	–	–	0.71	Ingrid↓	2.40e-46	95.0
Kaempferol-3-sophotrioside	4.70	773.2129	$[M + H]^+$	3	–	–	–	0.94	0.626	Ingrid↓	2.23e-21	77.8
9-OH-C18:3	7.62	277.2159	$[M + H]^+$	2, 3	0.356	18.3	928.4	0.816	0.311	Akooma/ Greenway↑	4.30e-26	83.7
C18:3	7.26	279.2315	$[M + H]^+$	2, 3	0.357	15.6	760.8	–	0.186	Akooma/ Greenway↑	2.31e-19	75.3
15,16-diOH-C18:2	7.38	295.2264	$[M + H]^+$	2	0.348	16.8	757.8	–	–	Akooma/ Greenway↑	3.53e-09	52.8
8-(3-octyl-2-oxiranyl) octanoic acid	7.68	281.2471	$[M + H]^+$	2	0.478	3.7	788.0	0.74	–	Akooma/ Greenway↑	1.41e-06	43.5
12(13)epoxy-C18:1	8.42	297.2422	$[M + H]^+$	2, 3	0.229	15.8	875.6	0.82	0.296	Akooma/ Greenway↑	1.03e-07	47.9
13-keto-C18:2	8.10	295.2266	$[M + H]^+$	2, 3	0.183	15.3	773.0	0.86	0.469	Akooma/ Greenway↑	7.52e-07	44.6
9,10-dihydrojasmonic acid	6.17	195.1375	$[M + H]^+$	2, 3	0.471	6.6	605.2	–	0.251	Akooma/ Greenway↑	1.90e-06	42.9
9-oxo-C18:2	6.18	295.2264	$[M + H]^+$	2, 3	0.335	16.5	610.4	–	0.426	Akooma/ Greenway↑	3.65e-11	58.2
Isobutyroylaspartic acid	4.55	218.1019	$[M + H]^+$	3	–	–	–	–	0.888	Kaplan/ Orchestra↑	7.48e-29	85.9
Isovaleroylglutamic acid	4.80	232.1176	$[M + H]^+$	2,3	–	–	–	0.737	0.943	Kaplan/ Orchestra↑	2.52e-22	79.1
Soyasaponin I	6.43	943.5257	$[M + H]^+$	3	–	–	–	0.943	0.937	Karacter↓	1.08e-11	59.9
Soyasapogenol B base + 3-O-GluA-Xyl-Rha, 22-O-Ara	6.34	1045.558	$[M + H]^+$	3	–	–	–	0.923	0.897	Akooma/ Greenway↑	1.52e-11	59.4
Bersimoside I	5.11	1267.633	$[M + H]^+$	3	–	–	–	–	0.735	Karacter↓	7.13e-09	51.4
oxIAA-Asp-N-Glc	4.21	469.1449	$[M + H]^+$	3	–	–	–	–	0.688	Akooma↓	9.60e-10	54.3

RT (min), retention time in minutes; *m/z*, experimental *m/z* value; Adduct, detected ion species; Annotation level, annotation confidence level, where Level 2 is a spectral library match and Level 3 is a computationally predicted structure. Δ*m/z*, mSigma, and MSMS score are parameters used for metabolites annotation using spectral libraries in Metaboscape showing accurate mass, isotopic score, and tandem mass spectra similarities, respectively. Cosine scores show fragmentation pattern similarity in GNPS molecular networking, confidence scores represent approximate COSMIC confidence scores assigned to structure identification in SIRIUS. *P*-values and effect sizes derive from one-way ANOVA with Benjamini-Hochberg correction for multiple testing (FDR = 5%) as well as arrows (↑/↓) indicating significantly higher or lower metabolite relative abundance.

lites with greater accumulation in Hammel samples (Fig. S3A). The pea phytoalexin pisatin grouped together with argininosuccinic acid, homocitrulline, asparagine, homoarginine, and spermidine in the second cluster. These compounds then clustered together with Arg-C5:0, which was most abundant in Sejet samples and slightly less abundant in Hammel samples.

3.4. Effect of cultivar

The cultivar effect explained 28.6% ($P = 0.001$) of the variation in pea seed metabolome data, consisting of 121 Level 2 identified metabolites, as determined by ASCA. The ASCA scores plot indicated a clear separation of Akooma and Greenway cultivars from the remaining eight cultivars along SC1 (Fig. 2C). The corresponding loadings plot suggested that these two cultivars contained relatively higher levels of fatty acids and their derivatives (Fig. 2D) specifically oxidation products of polyunsaturated fatty acids (C18:n), such as 9-oxo-octadecadienoic acid, 9-hydroxy-octadecatrienoic acid, octadecynoic (stearolic) acid, and 8,10-octadecadienal. The remaining eight cultivars were partially separated along SC2, though the location effect seemed to remain since replicates of cultivars were further separated from each other depending on the farming site.

The ASCA conducted on 178 metabolites identified using SIRIUS revealed a significant cultivar-influenced metabolic variation, accounting for 30% of the total variation ($P = 0.001$). The ASCA scores plot displayed a clear separation of Akooma, Greenway, and Ingrid cultivars, and partial separation of Orchestra, along

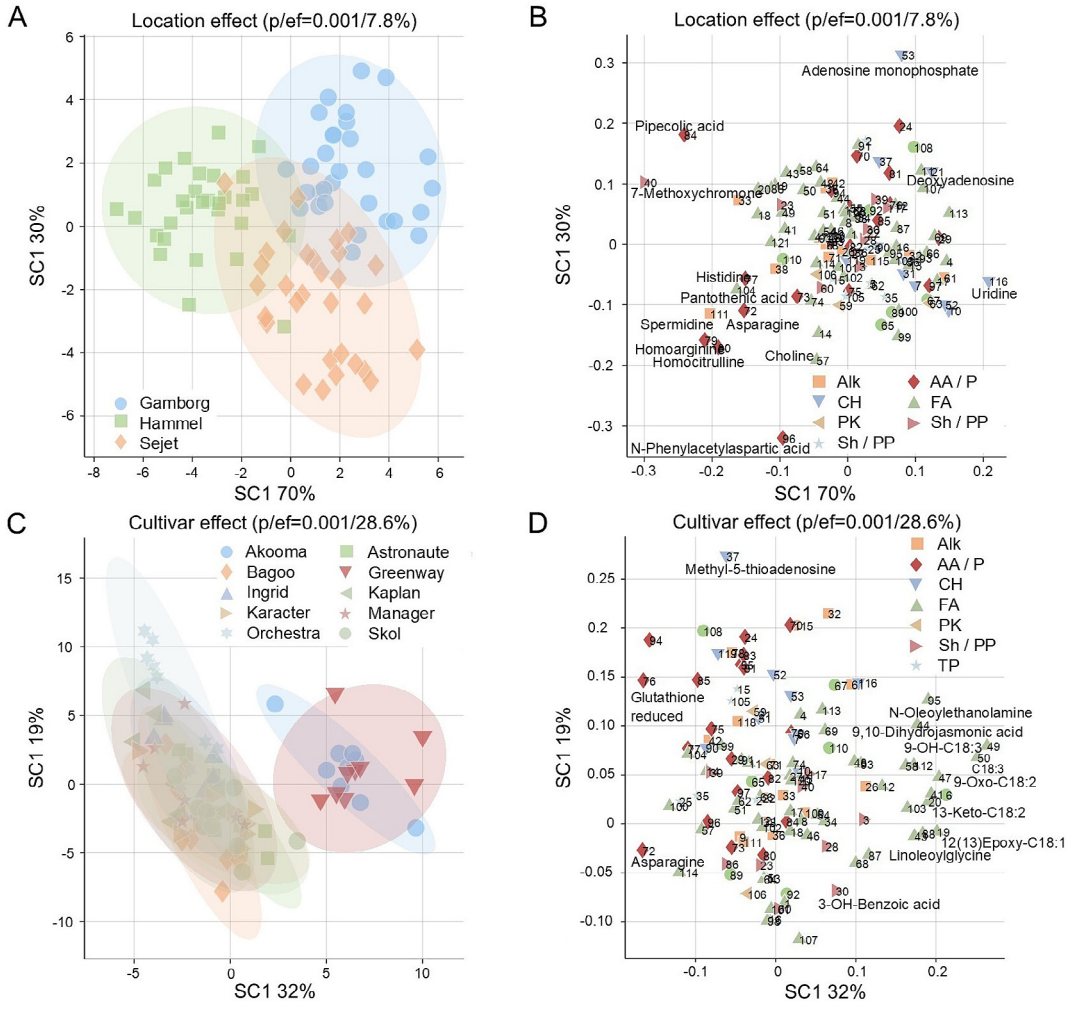


Fig. 2. ANOVA-simultaneous component analysis (ASCA)-based location and cultivar effect. (A) Scores plot demonstrates partial separation of pea samples according to the three locations (variance explained = 7.8%, *P*-value = 0.001) ($n = 3 \times 30$) with more prominent differentiation of pea grown at the organic farming site Hammel. Loadings plots show the distribution of metabolites, colored according to biosynthetic pathways (Alk, alkaloids; AA/P, amino acids and aeptides; CH, carbohydrates; FA, fatty acids; PK, polyketides; Sh/PP, shikimates and phenylpropanoids; TP, terpenoids) they derive from, explaining location (B) and cultivar (D) effects. (C) Scores plot highlights cultivar effect (variance explained = 28.6%, *P*-value = 0.001) for all pea samples from three locations ($n = 10 \times 9$) with Akooma and Greenway being most differentiated from other cultivars due to polyunsaturated fatty acids profile.

SC1 (Fig. S1C). These four cultivars were further differentiated from the remaining six cultivars along SC2. Notably, caffeoyl-, coumaroyl-, and feruloyl aspartates contributed to differentiation along SC2. Carboxylic acid conjugates of aspartate and glutamate (Asp-C4:0 and Glu-C5:0, respectively), indole-3-acetyl-N-aspartic acid (IAA-Asp), 9-oxo-octadecadienoic acid, 9-oxo-octadecatrienoic acid, and soyasaponin I were the main metabolites responsible for the separation along SC1.

The ASCA performed on the entire dataset, comprising 1230 features, demonstrated an even stronger separation among cultivars, which explained 19.2% of total variation ($P = 0.001$) (Fig. S2C). Although the overall variance explained by cultivar effect was lower in the full dataset compared to the annotated metabolite subset (both for Level 2 and 3 annotated), the ASCA scores plot revealed visually clearer separation among cultivars. This indicates that the inclusion of all detected features captured additional cultivar-specific patterns leading to more distinct discrimination. Notably, Orchestra, Kaplan, Ingrid, Manager, Bagoo, Skol, and Karacter cultivars formed two distinct populations along SC2, indicating an impact of location, which differentiates peas from the same cultivar. The ASCA loadings plot illustrates that Glu-C5:0, Asp-C4:0, benzoylaspartic acid, and dimethylarginine are the key drivers for this clustering pattern (Fig. S2D). Interestingly, Akooma and Greenway cultivars again formed a distinct cluster separated from the other cultivars.

One-way ANOVA revealed that 74 out of 121 Level 2 identified metabolites exhibited significant differences among pea cultivars, independent of the location effect. Chromone showed the highest effect (92.5%, $P = 4.9e\text{-}39$), being most abundant in Orchestra samples, followed by Karacter, and absent in Ingrid (Fig. 4A). Ingrid samples also displayed negligible levels of protocatechuic acid ($\eta^2 = 80.1\%$, $P = 9.80e\text{-}23$), a metabolite previously correlated with antioxidant capacity in peas [30]. Akooma and Greenway cultivars contained the highest levels of polyunsaturated fatty acid (PUFA)-derived oxylipins, including 9-hydroxy-octadecatrienoic acid ($\eta^2 = 83.7\%$, $P = 5.44e\text{-}26$), 9-oxo-octadecadienoic acid ($\eta = 76\%$, $P = 1.10e\text{-}19$), and octadecatrienoic acid ($\eta^2 = 75.3\%$, $P = 2.34e\text{-}19$). Regarding amino acids, Kaplan and Bagoo cultivars showed the highest abundance of aspartic acid ($\eta = 66.2\%$, $P = 2.76e\text{-}14$), while Ingrid exhibited the lowest. Kaplan samples were also notably richer in asparagine ($\eta^2 = 59.5\%$, $P = 2.35e\text{-}11$), with Greenway having the least abundance. Homocitrulline levels were lowest in Bagoo, Greenway, and Ingrid cultivars ($\eta = 52.6\%$, $P = 5.08e\text{-}09$). Although homocitrulline is associated with urea cycle disorders

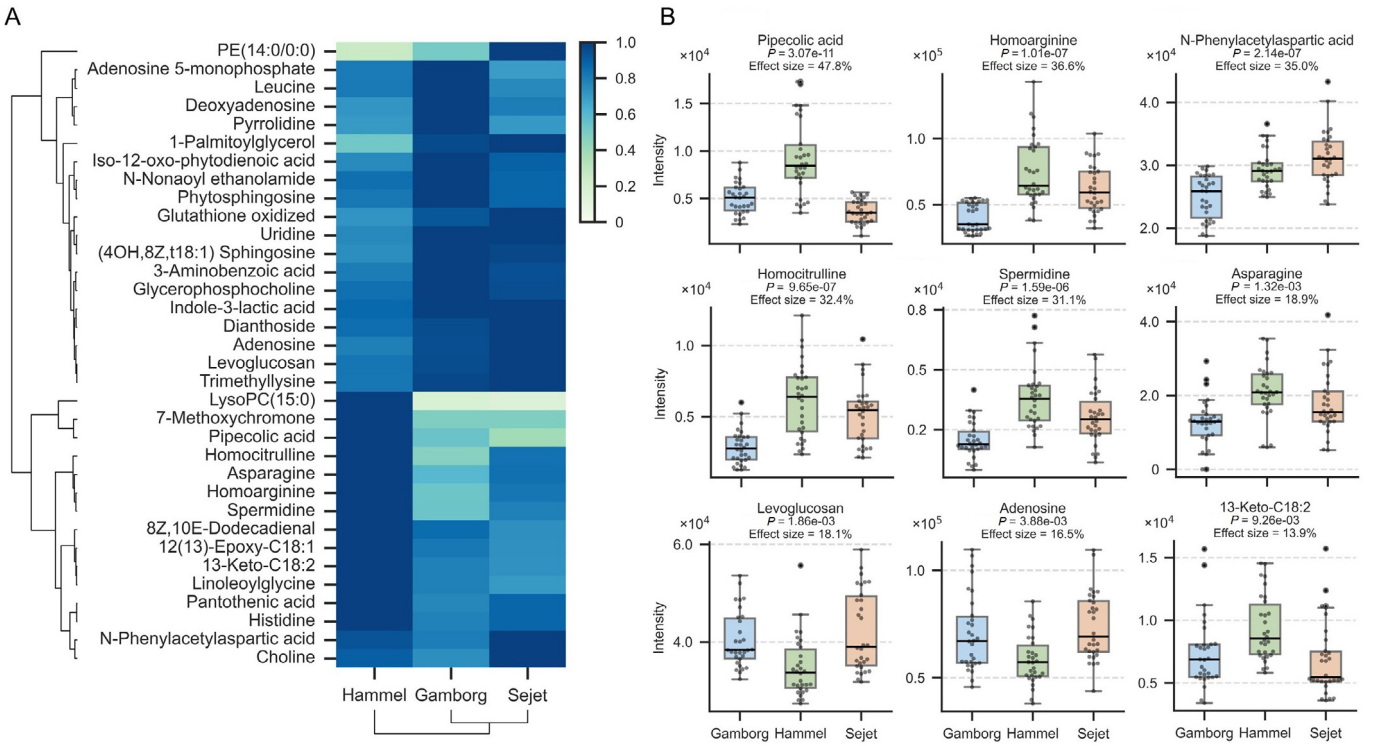


Fig. 3. Cultivar-independent variation of metabolite levels annotated by spectral libraries across three locations. (A) Hierarchically clustered heatmap of 34 metabolites identified by one-way ANOVA (FDR-adjusted *P*-values < 0.05 using Benjamini-Hochberg's correction for multiple testing) as significantly different between the three locations. (B) Box plots of selected metabolites with distinct relative abundance across locations identified as significant by one-way ANOVA.

in humans, its biological role in peas remains unexplored [31]. Leucine (η^2 = 36.4%, *P* = 6.62e-05) and 5-methylthioadenosine (η^2 = 58.6%, *P* = 5.08e-11), a metabolite involved in the plant methionine salvage pathway, were highest in Orchestra samples, with other cultivars showing comparable levels. Methionine (η^2 = 53.3%, *P* = 3.05e-09) also exhibited the highest abundance in Orchestra samples.

The relative abundances of all 74 significant metabolites influenced by cultivar effect are illustrated in a hierarchically clustered heatmap revealing three distinct metabolite clusters (Fig. 4B). The first cluster included lipid oxidation products and lipid-like compounds, predominantly abundant in Akooma and Greenway. The second cluster included metabolites with relatively uniform intensity patterns across cultivars, including metabolites palmitoylethanolamide, linoleoylethanolamide, tuberonic acid, and phytotoxin vomifoliol. The third cluster consisted of metabolites more abundant in Kaplan (asparagine, aspartic acid, and N-benzoylaspartic acid) and Orchestra (leucine, histidinol, methionine, and 5-methylthioadenosine).

Among the 178 SIRIUS-annotated metabolites, 105 exhibited significant differences attributed to cultivar effect. The three metabolites with the highest effect sizes were N-caffeoylaspartic acid (η^2 = 96.2%, *P* = 7.61e-51), N-feruloylaspartic acid (η = 95%, *P* = 2.40e-46), and N-coumaroylaspartic acid (η^2 = 93.7%, *P* = 2.54e-42) (Fig. S4B). These metabolites showed nearly identical patterns across cultivars, being most abundant in Orchestra and Karacter samples and absent in Ingrid samples. Isobutyryl aspartic acid (η^2 = 85.9%, *P* = 7.48e-29) exhibited the highest abundance in Orchestra, Kaplan, and Ingrid samples and the lowest in Manager, Karacter, and Bagoo samples. Soyasaponin I (η^2 = 59.9%, *P* = 1.09e-11), one of the most abundant triterpenoid saponins identified in peas, showed the highest levels in Greenway, followed by Bagoo, Orchestra, Karacter, Skol, and Akooma samples. Conversely, bersimoside I (η^2 = 51.4%, *P* = 7.21e-09) and pisumsaponin I (η^2 = 42.2%, *P* = 2.61e-06) displayed, in total, the lowest abundance in pea samples. Another saponin annotated as soyasapogenol B base-3-O-GluA-Xyl-Rha-22-O-Ara (η^2 = 59.4%, *P* = 1.54e-11) had higher concentrations in Akooma and Greenway cultivars. Dimethylarginine (η^2 = 90.9%, *P* = 2.82e-36) was predominantly found in Akooma, Ingrid, and Greenway samples, with lower levels in other cultivars. The highest level of isovaleryl glutamic acid (η^2 = 79.1%, *P* = 2.52e-22) was in Kaplan, followed by Orchestra and Ingrid, and absent in Bagoo and Karacter samples.

Hierarchical clustering of the SIRIUS-annotated 178 metabolites identified three distinctive clusters (Fig. S4A). One cluster consisted of conjugates of aspartic and glutamic acids with cinnamic acid derivatives and short-chain fatty acids, predominantly abundant in Orchestra, Kaplan, and Karacter cultivars. Ingrid samples were devoid of hydroxycinnamic acid aspartates regardless of location type. A second cluster, characterized by lipid oxidation products, exhibited the highest intensities in Akooma and Greenway samples, and included the triterpenoid saponin soyasapogenol B base-3-O-GluA-Xyl-Rha-22-O-Ara. In the third major cluster, triterpenoid saponins soyasaponin I and pisumsaponin I as well as jasmonates 4,5-didehydrojasmonic acid and tuberonic acid clustered with one another.

3.5. Tentative identification of novel indole metabolites and triterpenoid saponins in pea seeds using mass spectral computational approaches

Combining molecular networking-based annotation in GNPS with computational methods in SIRIUS enabled the identification of additional metabolites significantly influenced by location and cultivar effects. Specifically, a metabolite with a precursor *m/z* value of 469.145, annotated by SIRIUS as 1-N-β-glucopyranosyl-2-oxindole-3-acetyl-N-aspartic acid (oxIAA-Asp-N-Glc), showed significant variability across cultivars (η^2 = 54.3%, *P* = 9.6e-10)

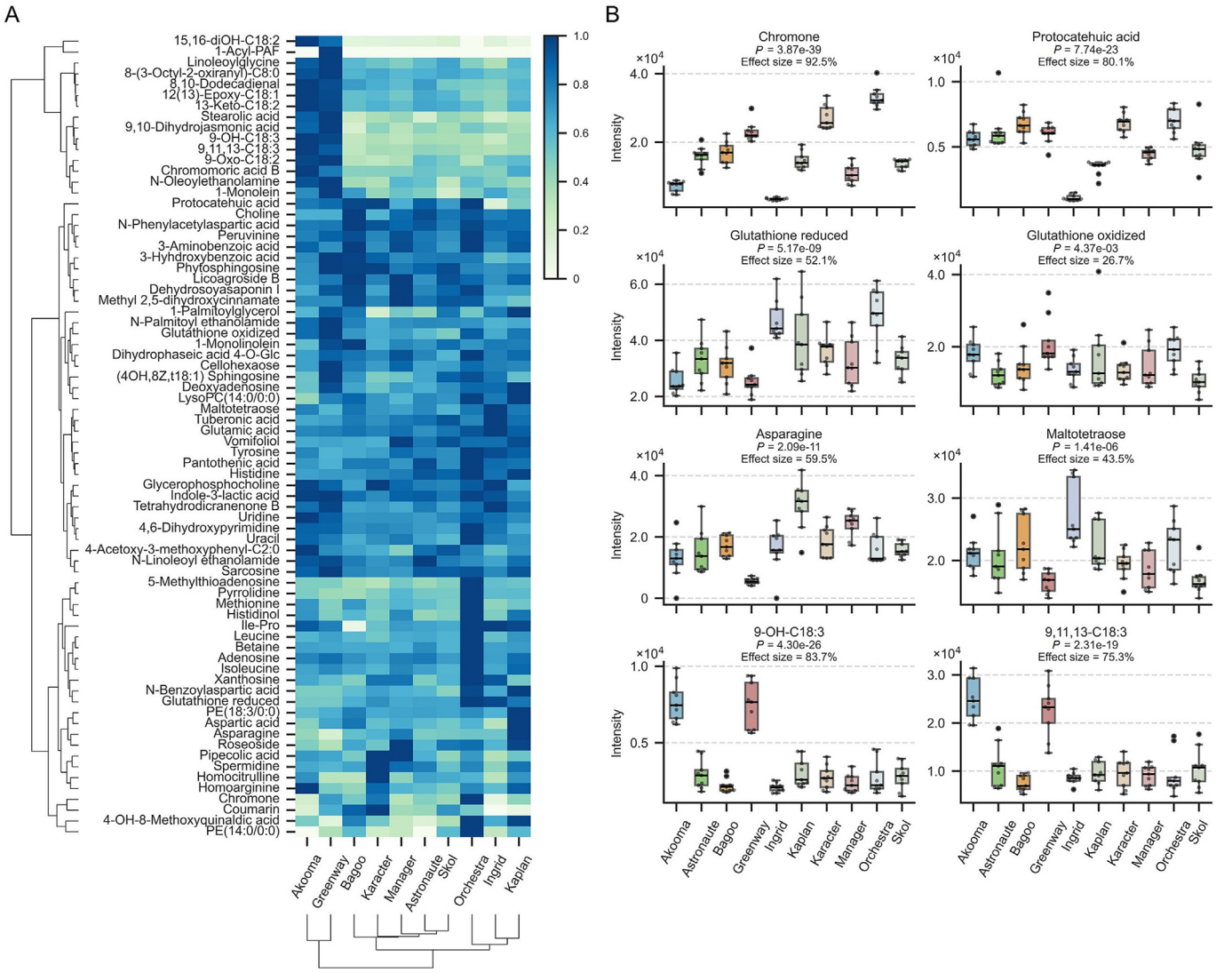


Fig. 4. Location-independent variation of metabolite levels annotated by spectral libraries across ten cultivars. (A) Hierarchically clustered heatmap of 74 metabolites identified by one-way ANOVA (FDR-adjusted *P*-values < 0.05 using Benjamini-Hochberg's correction for multiple testing) as significantly different between the cultivars. (B) Box plots of selected metabolites with distinct relative abundance across cultivars identified as significant by one-way ANOVA.

(Fig. S4A, B). Although this metabolite was not identified using spectral libraries employed in our study, molecular networking provided meaningful chemical relationships between this metabolite and five other features (Fig. 5A). OxIAA-Asp-N-Glc was linked to other features through mass differences (Δ *m*/*z*) of 162.053, 15.995, and 33.961 Da. The first two delta masses correspond to glycosylation and hydroxylation, respectively. Literature data confirms that oxIAA-Asp-N-Glc can be further glycosylated to form oxIAA-Asp-N-Glc-Glc (*m*/*z* 631.1975) and these metabolites were initially identified in tomato pericarp tissue [32]. The experimental fragmentation patterns from our study matched literature data for all these features. Consecutive glucose losses from precursor ion with *m*/*z* 631.1975 yielded ions at *m*/*z* 469.1449 and 307.0923. Additional diagnostic fragment ions for oxIAA-Asp moiety were *m*/*z* 216.065 and 188.07, arising from cross-ring cleavage within the glucose ring combined with β-cleavage of the amide bond and further loss of CO, respectively. Next characteristic fragments were observed at *m*/*z* 174.054, resulting from sugar cleavage from the indole moiety combined with β-cleavage of the amide bond, and at *m*/*z* 146.059, a quinolonium fragment from the indole structure.

Regarding hydroxylation products, oxindole-3-acetic acid has been previously reported to undergo oxidation, forming 3-hydroxy-2-oxindole-3-acetic acid [33]. This compound is subsequently glycosylated at the hydroxyl group to yield 3-O-β-glucopyranosyl-2-oxindole-3-acetyl-N-aspartic acid (dioxIAA-Asp-3-Glc, *m*/*z* 485.13), which was first identified in faba bean seedlings (*Vicia faba* L. cv. Chukyo), with detailed spectral characterization reported [34]. Given the absence of mass spectral evidence, specifying the precise hydroxyl group position and additional hydroxylated analogs of oxIAA-Asp-N-Glc, the metabolite at *m*/*z* 485.1396 was tentatively annotated as dioxIAA-Asp-3-Glc.

The mass difference of 33.961 Da often indicates the presence of chlorine, which we confirmed from the isotopic pattern of the precursor ion at *m*/*z* 503.1056 exhibiting characteristic isotopic peaks for the two most abundant natural chlorine isotopes, ^{35}Cl and ^{37}Cl. Specifically, alongside the precursor ion at *m*/*z* 503.1056 (2.3% intensity), an isotopic peak at *m*/*z* 505.1071 (1.0% intensity) was observed. Further, neutral loss of a hexose moiety (−162 Da) generated fragment ions at *m*/*z* 341.0525 (35.3%), differing by 33.961 Da from similar glucose-loss fragments from oxIAA-Asp-N-Glc, and at *m*/*z* 343.0382 (8.8%). Likewise, all other characteristic

oxIAA fragment ions shifted to 33.961 Da and displayed chlorine isotope patterns, confirming chlorine substitution within the indole moiety (Fig. 5B). Therefore, the metabolite with *m*/*z* 503.1056 was tentatively annotated as 4-chloro-1-N-β-glucopyranosyl-2-oxindole-3-acetyl-N-aspartic acid (4-Cl-oxIAA-Asp-N-Glc). Molecular networking further linked this metabolite to features with mass differences of 162.053 and 15.99 Da. Examination of MS2 spectra allowed annotation of these metabolites as 4-Cl-oxIAA-Asp-N-Glc-Glc (*m*/*z* 665.1585) and 4-Cl-dioxIAA-Asp-3-Glc (*m*/*z* 519.1007). To our knowledge, these chlorinated derivatives of oxindole-3-acetic acid have not previously been reported.

4-Cl-oxIAA-Asp-N-Glc-Glc was the only chlorinated metabolite influenced by location (η = 24.5%, *P* = 7.22e-05), being more abundant in Hammel and Sejet than in Gamborg (Fig. S5A). All chlorinated and non-chlorinated oxIAA-Asp analogues were shown to be influenced by cultivar type in one-way ANOVA (Fig. S5B).

Similarly, metabolites annotated through SIRIUS as significant across locations and cultivars were further verified via molecular networking. Triterpenoid saponins proposed by SIRIUS lacked annotation from conventional spectral libraries, although some were tentatively dereplicated using a suspect library. For example, the metabolite with *m*/*z* 1069.558 was annotated by SIRIUS as soyasaponin βg (soyasapogenol B base + O-HexA-Hex-dHex, O-DDMP) and by the suspect library as suspect to soyasapogenol B base + O-HexA-Pen-dHex, O-DDMP (*m*/*z* 1039.547), differing by 30.011 Da (Fig. 6A). This mass difference corresponded to the –CH2O difference between pentose and hexose sugars, confirmed through MS2 spectral analysis showing pentose (–132 Da) and hexose (–162 Da) losses in reference and suspect spectra. Consequently, the metabolite at *m*/*z* 1069.558 was tentatively annotated as soyasaponin βg. This metabolite was connected to a feature at *m*/*z* 1231.601, not annotated by spectral libraries or SIRIUS, yet significantly influenced by location (η^2 = 23.5%, *P* = 1.25e-04) and cultivar type (η^2 = 46.9%, *P* = 3.72e-07). The 162.053 Da mass difference indicated an additional hexose unit, confirmed by MS2 spectra, revealing all characteristic soyasaponin βg ions following

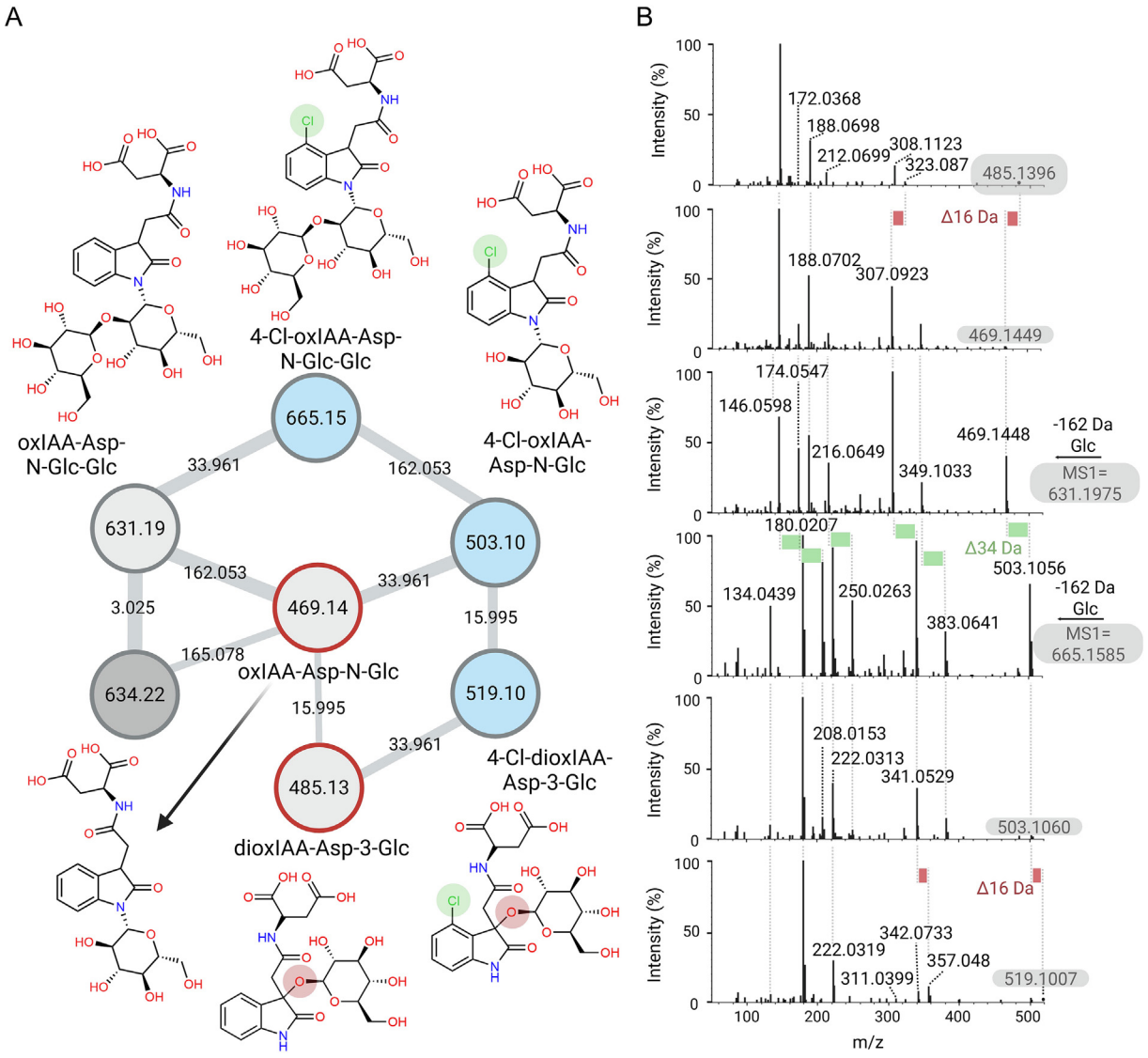


Fig. 5. Molecular networking and SIRIUS-assisted annotation of 2-oxindole-3-acetyl aspartic acid (oxIAA-Asp) molecular family. (A) The light grey nodes indicate the MS/MS spectra of candidates that were not annotated by spectral libraries, but matched with previously reported compounds in the literature, the dark grey nodes – not annotated. The blue nodes mean that the putative candidates have not been annotated or reported in literature. The nodes with red borders had matches with SIRIUS annotation (B) Tandem mass spectra show diagnostic peaks of putatively annotated compounds and shifts explaining the addition of –OH and –Cl.

precursor and hexose unit losses (Fig. 6). Based on available evidence, no previous studies have reported this glycosylated analog of soyasaponin βg.

4. Discussion

Pea (*Pisum sativum* L.) seed biochemistry integrates unique insights relevant to plant biology, nutrition, and global food sustainability. Therefore, metabolite profiling informs breeding and processing strategies to mitigate antinutritional factors and enhance sensory quality, with implications for both resilient agriculture and human health [6]. In this study, untargeted metabolomics was performed on ten pea cultivars grown at three geographically distinct sites in Denmark. The Hammel location practiced organic farming, whereas Gamborg and Sejet used conventional methods. It was found that Hammel-grown samples contained elevated levels of nitrogen-rich metabolites and non-proteinogenic amino acids, such as pipecolic acid, homocitrulline, asparagine, homoarginine, spermidine, argininosuccinic acid, and homopipecolic acid compared to those from Sejet and Gamborg. Despite the broader inference of this trend to a specific agricultural practices is limited by the design of present study, and consistent cultivar-independent observations support the association with organic farming. A critical distinction between organic and conventional farming systems is the source and timing of nitrogen availability. Conventional systems typically supply readily available inorganic nitrogen, whereas organic systems rely on microbial mineralization of organic matter, manure, compost, and nitrogen fixation by legumes. Additionally, organic practices promote diverse soil microbiomes, enhancing symbiotic relationships with rhizobia bacteria that convert atmospheric nitrogen (N_2) into ammonia (NH_3). However, excess ammonia is toxic, leading plants to rapidly assimilate ammonia into organic nitrogen-storing compounds. Generally, higher ammonium nutrition results in greater accumulation of free amino acids compared to nitrate-based nutrition [35]. Asparagine, for instance, functions as a primary nitrogen storage and transport amino acid in peas [35]. In pearl millet, asparagine was reported to mitigate ammonia toxicity during drought stress, serving as a nitrogen reserve that could be remobilized upon recovery of growth conditions [36]. Additionally, Igbasan et al. demonstrated that elevated nitrogen availability significantly increases arginine content in peas [37]. Under conditions of drought, extreme temperatures, or other abiotic stresses, plants commonly accumulate polyamines such as spermidine, which function as antioxidants and osmoprotectants [38]. Pipecolic acid, a lysine metabolism product, is recognized as a signaling molecule involved in plant immunity, notably accumulating during the activation of systemic acquired resistance (SAR) in response to pathogen attack [39,40]. As organic cultivation often subjects plants to greater stress, these findings may explain the elevated accumulation of nitrogen-rich amino acids and polyamines such as spermidine in peas grown under organic conditions in Hammel.

Compared to other cultivars, Akooma and Greenway pea cultivars exhibited notably higher levels of lipid peroxidation products, likely attributable to their distinct fatty acid compositions and enzyme activities. Pea seeds primarily contain linoleic acid (C18:2) and linolenic acid (C18:3). Oxidation of these polyunsaturated fatty acids (PUFAs) by the enzyme lipoxygenase (LOX) generates aldehydes, ketones, and hydroxy-fatty acids, collectively termed oxylipins. Oxylipins are signaling molecules in plants that mediate stress responses and innate immunity. These compounds are also known contributors to the characteristic “beany” off-flavor in peas [41]. Indeed, previous research linked the levels of volatile beany compounds to both PUFA content and LOX enzyme activity.

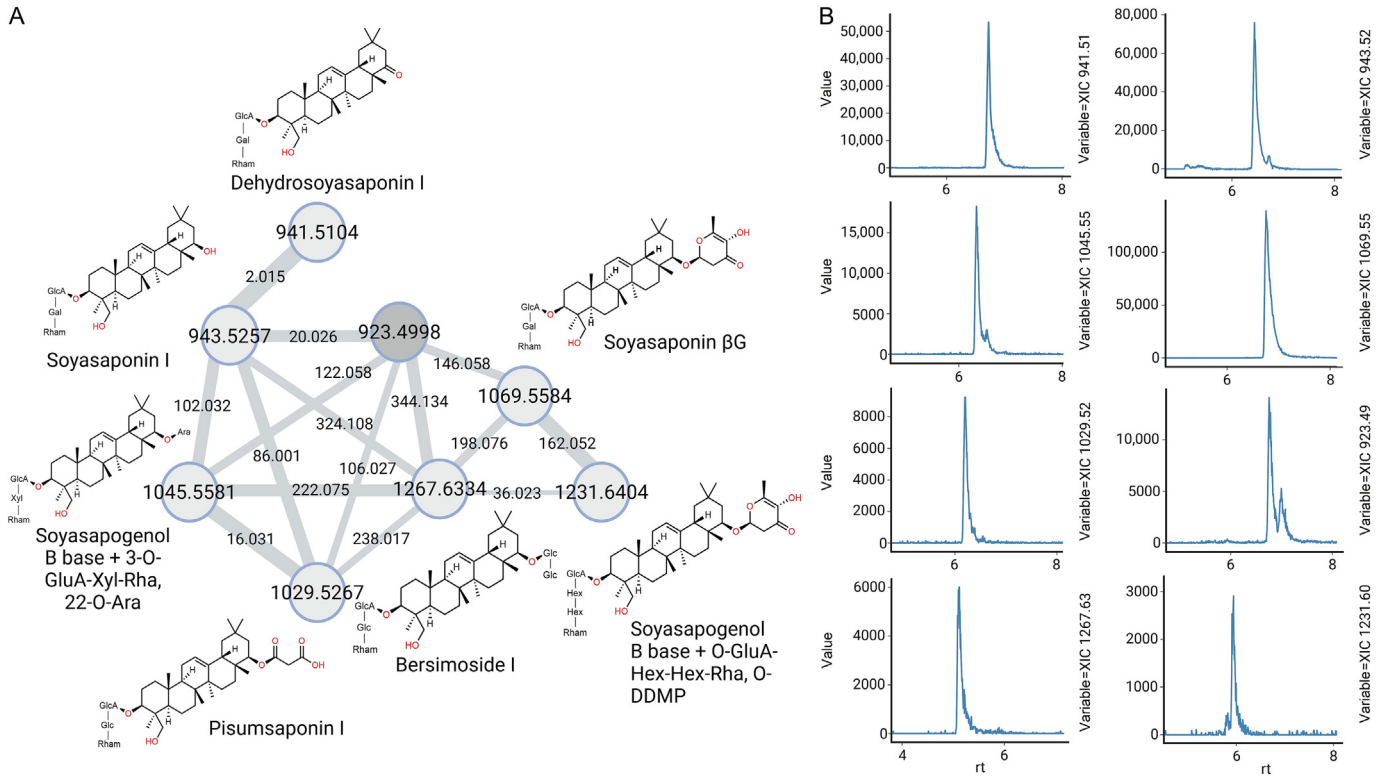


Fig. 6. Pea triterpenoid saponins. (A) Molecular networking of pea triterpenoid saponins (dark grey node indicates in-source fragment ion). Spectral libraries implemented in this study lacked reference data, except for dehydrosoyasaponin I, and were unable to annotate other triterpenoid saponins. Only the combination of suspect library hits confirmed with SIRIUS annotations and manual verification of fragmentation patterns allowed for their annotations. (B) Extracted ion chromatograms that correspond to detected triterpenoid saponins.

Consistent with these findings, linolenic acid was found in significantly greater abundance in Akooma and Greenway samples than in other cultivars (Fig. 4A, B), possibly providing increased substrate availability for LOX-mediated oxidation. Moreover, the higher levels of 13-keto-C18:2, epoxy-C18:1, and 11-OH-C18:2 observed in Hammel samples may indicate an important role of LOX-derived oxylipins in mediating plant-pathogen interactions under organic farming conditions [42].

Across all cultivars, soyasaponin βg was the predominant triterpenoid saponin, with soyasaponin I being the second most abundant (Fig. S4B). Soyasaponin βg is the DDMP-conjugated form of group B saponins to which soyasaponin I belongs. Several studies have shown that the DDMP moiety is chemically labile, readily cleaving to produce non-DDMP saponin such as soyasaponin I under thermal or oxidative conditions [43]. From a sensory perspective, DDMP-type saponins are consistently reported to contribute more strongly to bitterness and astringency than non-DDMP forms in peas and other legumes [8]. Thus, the dominant soyasaponin βg profile suggests that all cultivars in our study possess similar intrinsic bitterness potential. In peas, triterpenoid saponins as a class have been linked with plant defense mechanisms, including antifeedant functions and associations with partial resistance to root pathogens [44,45].

The phenylpropanoid pathway, particularly the biosynthesis of hydroxycinnamic acid amides (HCAAs), was also differentially influenced by cultivar type. Interestingly, pea cultivar Ingrid, which is popular in Northern Europe, exhibited complete absence or only negligible levels of aspartic acid conjugates of caffeic, ferulic, and coumaric acids along with chromone, kaempferol-3-sophotrioside, and protocatechuic acid (3,4-dihydroxybenzoic acid) that are also part of the phenylpropanoid pathway. This pattern was observed across all three field locations practicing both conventional and organic farming, while the same metabolites were readily detected in all other cultivars grown under identical conditions. In sensory terms, these HCAAs negatively correlate with bitterness, although their unconjugated forms, such as caffeic acid, positively correlate with bitterness perception in peas [46]. Cultivars with lower or undetectable amounts of phenylpropanoid pathway metabolites contribute less to the overall antioxidant potential of peas as demonstrated in comparative studies where differences in phenolic acid and flavonoid abundance were associated with reduced antioxidant activity across pea varieties [30]. A recent study has shown that wild pea genotypes have higher expression of enzymes involved in the phenylpropanoid pathway than domesticated varieties, namely COMT (caffeic acid 3-O-methyltransferase), which is responsible for conversion of caffeic acid to ferulic acid [47]. Overall, transcriptomic and metabolomic results in this study demonstrated altered profiles of expressed genes and metabolites, including hydroxycinnamic acids, favoring seed defense in wild (and pigmented) genotypes. Such modification may potentially reflect targeted breeding efforts aimed at improving taste, appearance or seed composition traits. This provides a plausible biological context for the absence, or markedly low levels of caffeoyl, feruloyl, and coumaroyl aspartates along with other phenylpropanoid pathway metabolites in the Ingrid cultivar. However, in the absence of genetic or transcriptomic data, it is impossible to determine whether this pattern reflects altered pathway regulation or environmentally modulated suppression. Moreover, because the experimental design of our study is not sufficient to assess any phenotypic traits, the functional consequences of such a metabolic pattern remain unknown and should be investigated in future work. Additionally, the specific physiological purpose behind the conjugation of these three hydroxycinnamic acids with aspartate has not been clearly elucidated to date. It is known that the acyl acid amido synthetases of the *Gretchen Hagen 3* (GH3) family conjugate amino acids, including aspartate, to phytohormones such as indole-3-acetic acid, salicylic acid, phenylacetic acid, and benzoic acids to regulate levels of their active and inactive forms [48]. But whether GH3 enzymes could conjugate hydroxycinnamic acids to aspartate for similar purposes is yet to be determined.

Furthermore, 1-N-β-glucopyranosyl-2-oxindole-3-acetyl-N-aspartic acid (oxIAA-Asp-N-Glc), annotated by SIRIUS, displayed significant cultivar-dependent variability. Expansion of this annotation through molecular networking uncovered three novel chloroauxin metabolites, which were tentatively annotated by manual interpretation of their MS/MS spectra. Typically, structural elucidation of previously unknown compounds requires detailed nuclear magnetic resonance (NMR) and mass spectrometry analysis of pure, isolated, or synthesized reference compounds. Subsequent matching of spectral data and retention times from biological samples confirms the presence of candidate metabolites. Although the chloroauxins proposed here lack direct verification through such approaches, their annotations are robustly supported by well-characterized fragmentation patterns of non-chlorinated analogues and distinct chlorine isotopic signatures. Additionally, a plant-specific mass spectral repository search (plantMASST) confirmed the presence of 4-Cl-oxIAA-Asp-N-Glc in publicly available pea metabolomics data (Fig. S6), affirming its occurrence as a common pea metabolite [49]. In fact, a previous pea metabolomics study also reported two metabolites at Level 3 identification as 'chlorinated aspartic acid derivatives', the precursor masses of which match 4-Cl-oxIAA-Asp-N-Glc and 4-Cl-dioxIAA-Asp-3-Glc found in our study [50]. These findings emphasize the significant utility of computational tools and molecular networking in uncovering previously uncharacterized plant metabolites.

The proposed chloroauxin metabolites are analogues of indole-3-acetic acid (IAA), a widely occurring phytohormone involved in plant growth. Recently, comprehensive insights into the biosynthesis of IAA and its reversible and irreversible inactivation pathways have been described [51]. IAA homeostasis is maintained through reversible inactivation by conjugation with amino acids and sugars (via *GH3-IAA-Leu-Resistant1 (ILR1)* enzymes and UDP-glycosyltransferases, respectively) and irreversible catabolism via oxidation of the indole moiety (via *DIOXYGENASE FOR AUXIN OXIDATION 1 (DAO1)-ILR1)* [33]. Notably, several agriculturally significant legumes, including pea, lentil, and faba bean, produce the more potent auxin analogue 4-chloroindole-3-acetic acid (4-Cl-IAA), which plays an essential role in pod elongation [52]. However, the metabolic fate of 4-Cl-IAA – whether it undergoes degradation or conjugation – remains unclear [53]. Previous studies have identified 4-Cl-IAA methyl ester and its aspartate amide conjugate in pea seeds [54], and quantitative analyses have detected these compounds in young developing seeds and pericarps [55]. Moreover, a recent study demonstrated *in vitro* that chlorinated IAAs can be converted to amino acid conjugates by the GH3 auxin amino acid synthetase family, enzymes known for their role in IAA-amino acid conjugation [56]. These findings suggest that 4-Cl-IAA catabolism may closely resemble the metabolic pathways of non-chlorinated IAA, involving amino acid conjugation. However, oxidative catabolism of 4-Cl-IAA has not yet been reported, and further research is necessary to elucidate the enzymes involved.

Although the current design provided a valuable perspective by assessing location- and cultivar-dependent metabolic variation of pea seeds, several limitations should be acknowledged. Organic farming was represented by a single site, implying that agricultural practices and environmental factors (e.g., soil composition, climate) cannot be fully separated. Therefore, the observed differences between organically grown Hammel samples and conventional sites Gamborg and Sejet should be interpreted only as associations with organic practice. Future studies where both organic and conventional farming practices are implemented

across multiple locations would be required to distinguish the farming practice effects. Nevertheless, the observed cultivar-independent trends in this study provide an understanding of possible metabolic responses of pea seeds to different growing environments.

In summary, this study illustrates the impact of growth conditions, specifically conventional versus organic farming practices, and cultivar differences in the pea seed metabolome. Both multivariate and univariate analyses demonstrated that selected pea cultivars exhibited biochemical adaptations to environmental and genetic factors via alterations in amino acid, fatty acid, phenylpropanoid, and triterpenoid saponin pathways. Furthermore, variations in compounds associated with sensory off-flavors were examined. Enhancing traditional untargeted metabolomics approaches with computational annotation methods significantly expanded our understanding of location- and cultivar-specific metabolites and facilitated the identification of novel chloroauxin catabolites.

CRediT authorship contribution statement

Sarvar A. Kakhkhorov: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Søren Balling Engelsen:** Writing – review & editing, Conceptualization. **Kristian Holst Laursen:** Writing – review & editing, Conceptualization. **Bekzod Khakimov:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Conceptualization.

Data availability

Supplementary data for this article can be accessed at the Science Data Bank by https://doi.org/10.57760/sciencedb.j00210.00111 as well as through https://github.com/BEKZODKHAKIMOV/LC-MS-based-Pea-Metabolomics. Untargeted LC-MS/MS data generated for this study are publicly available both at Global Natural Products Social Molecular Networking (GNPS)/MassIVE (https://massive.ucsd.edu/) under the accession code MSV000100511 and the Science Data Bank by https://doi.org/10.57760/sciencedb.j00210.00111. Feature-based molecular networking job can be accessed at: gnps2.org/status?task=a4c1ca4f1f8e4312b9204e23948bee6f.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data for this article can be found online at https://doi.org/10.1016/j.cj.2026.01.012.

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